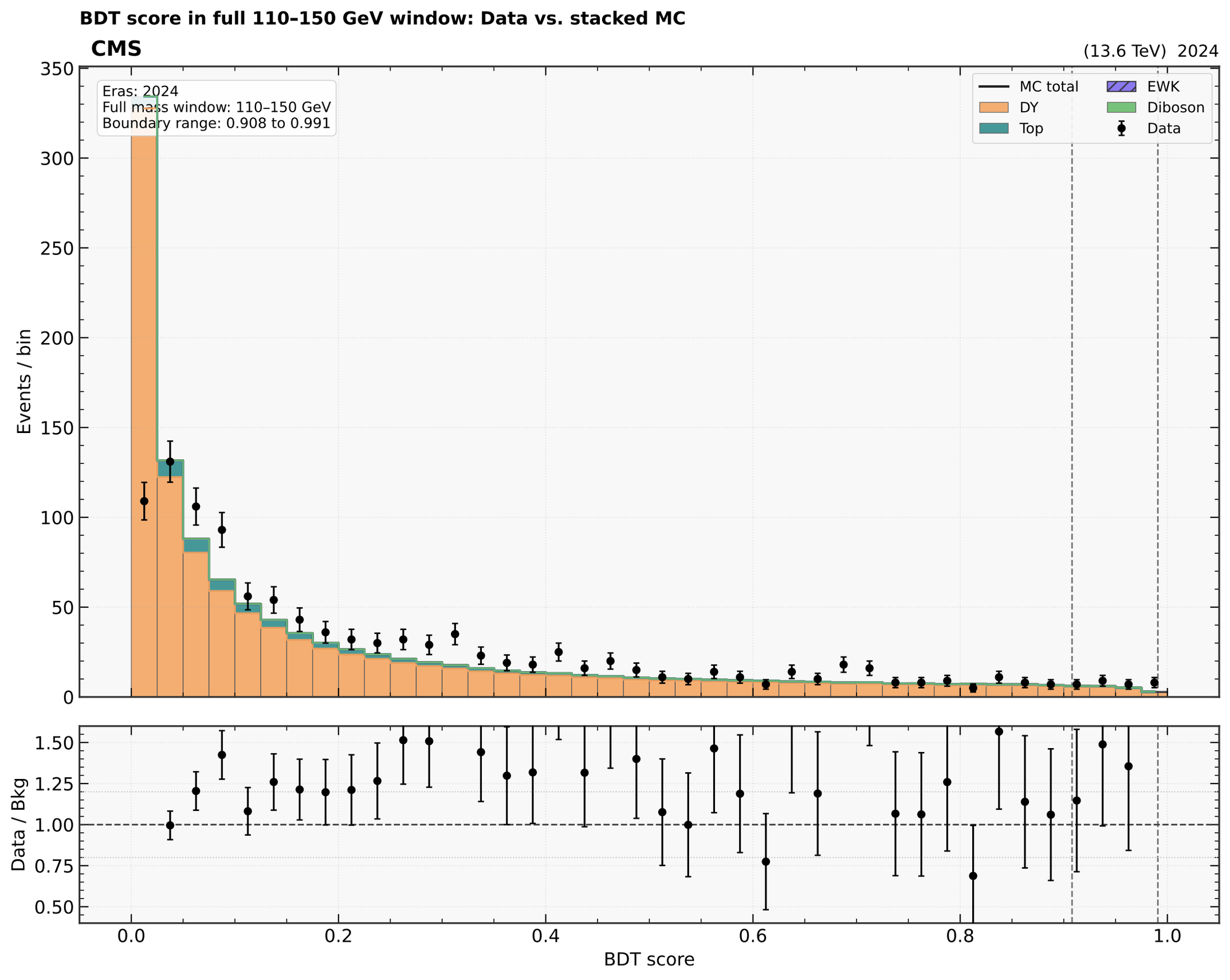
Run-3 H → μμ VBF BDT Category Optimisation

Eras : 2024
Category edges : 0.0000, 0.9080, 0.9837, 0.9908, 1.0000
MC scale factor : 1.0000 (Data/MC sideband normalisation)
Blinding window : 115–135 GeV
Sidebands : (110,115) ∪ (135,150) GeV
Outputs dir : /eos/home-m/memmad/HiggstoDimuon_VBF/src/drivers/9_TMVA/Outputs
Extra selection : –

CONTENTS:

p.2 BDT score distribution: Data vs. MC
p.3 BDT bin yields
p.4 Significance vs. number of categories
p.5 Boundary scan step-by-step (AN Fig. 32)
p.6 Atanh BDT + S/√B per category
p.7 BDT output vs. mass hypothesis (AN Fig. 26R)
p.8 Signal FWHM vs. BDT score
p.9 ROC curve with AUC
p.10 Blinded dimuon mass spectrum
p.11 Mass sculpting check (AN Fig. 25)
p.12 Per-category mass distributions and background fits
p.13 Background model comparison (χ²/ndf)
p.14 VBF signal DCB fits per category
p.15 DCB parameter table
p.16 σ_CB vs. category
p.17 Signal acceptance × efficiency
p.18 Per-category yield and sensitivity summary
p.19 Signal mass shapes mH=120/125/130 (AN Fig. 40)
p.20 Category summary table
p.21 Category process composition

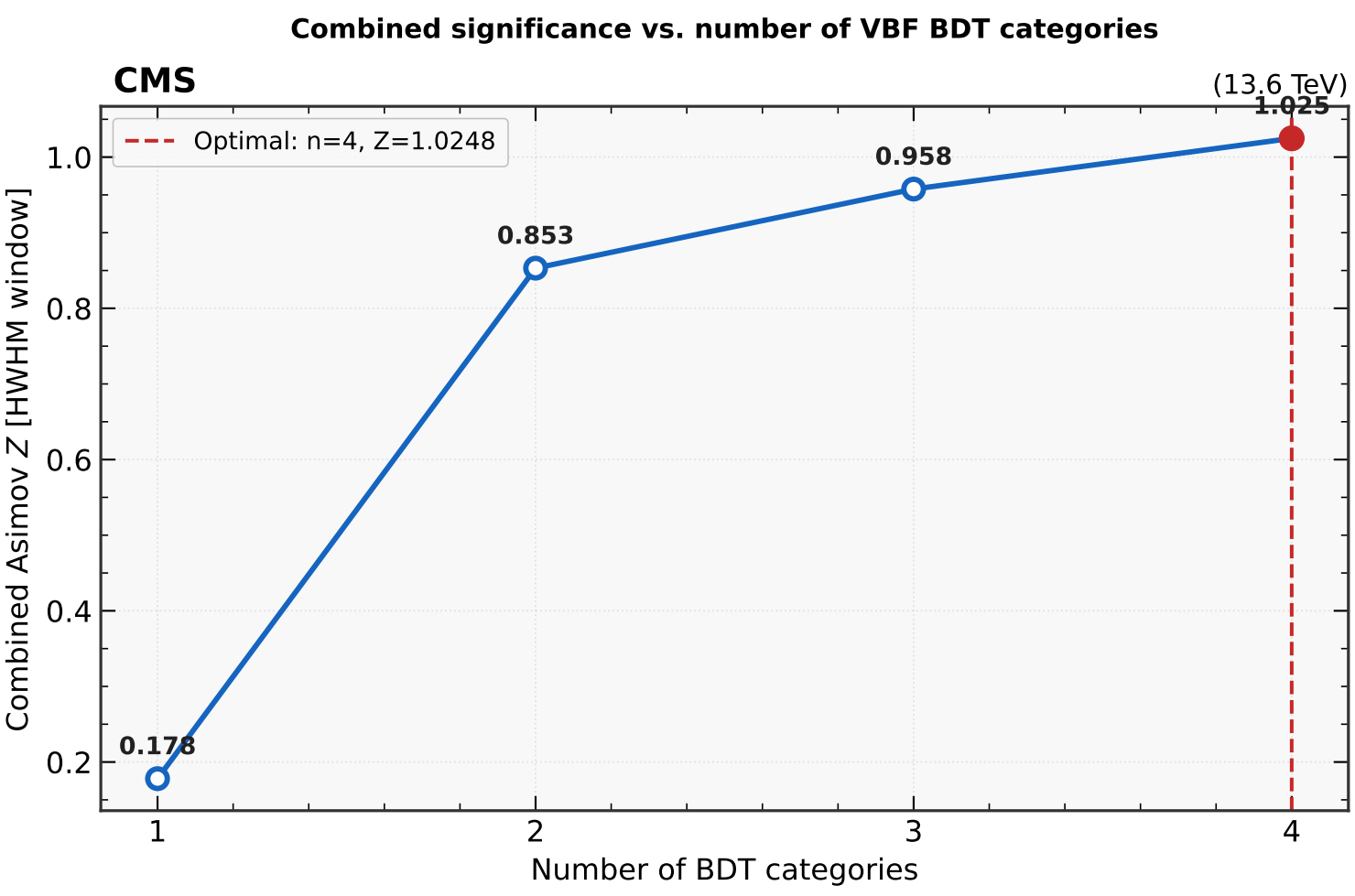

BDT score in full 110–150 GeV window: Data vs. stacked MC
CMS
(13.6 TeV) 2024
Eras: 2024
Full mass window: 110–150 GeV
Boundary range: 0.908 to 0.991
MC total
DY
Top
EWK
Diboson
Data
Events / bin
Data / Bkg
BDT score
0
50
100
150
200
250
300
350
0.50
0.75
1.00
1.25
1.50
0.0
0.2
0.4
0.6
0.8
1.0

BDT bin summary: signal, background, and sensitivity (page 1 / 2)

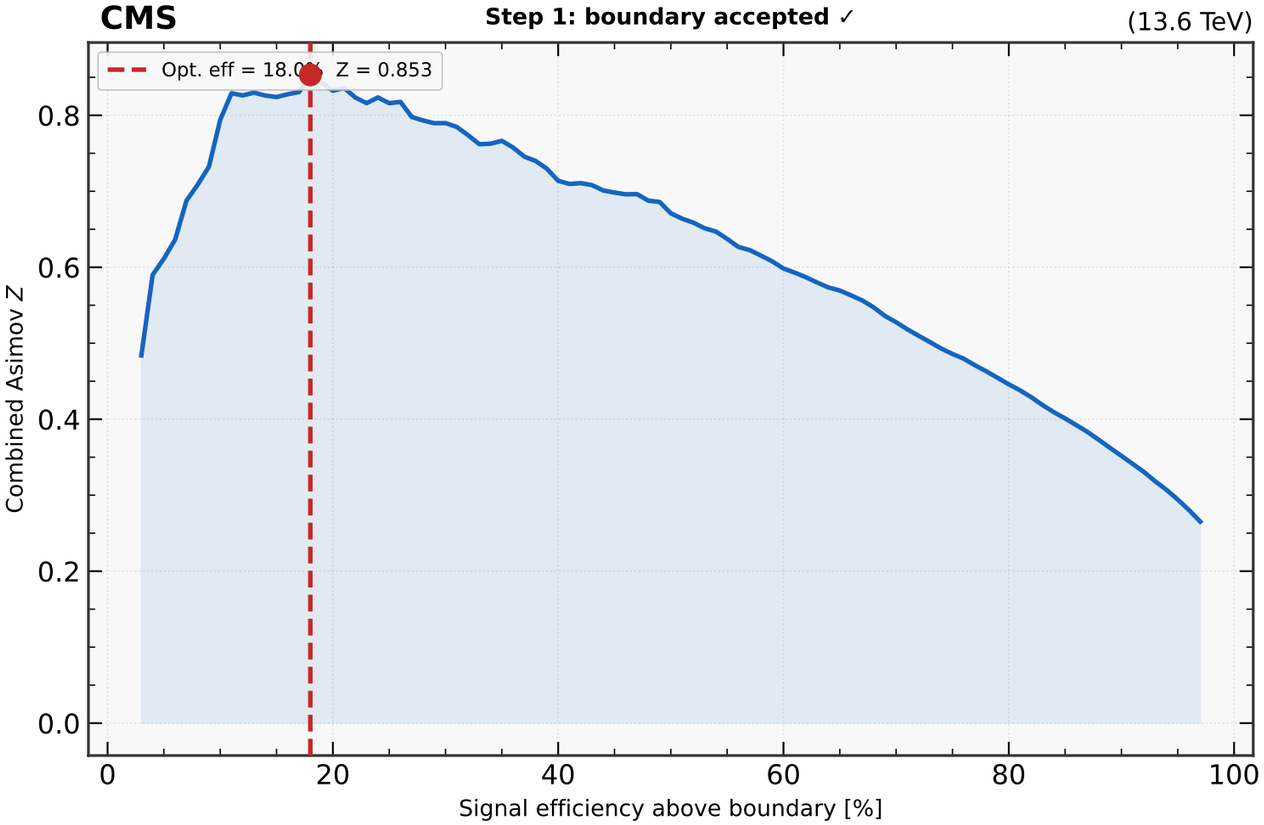
BDT bin	Total signal	Bkg. ± ΔB	Data	S/(S+B) (%)
[0.000, 0.025)	0.171	334.31 ± 1.03	109	0.05
[0.025, 0.050)	0.250	131.59 ± 0.63	131	0.19
[0.050, 0.075)	0.281	87.98 ± 0.51	106	0.32
[0.075, 0.100)	0.289	65.29 ± 0.44	93	0.44
[0.100, 0.125)	0.295	51.78 ± 0.39	56	0.57
[0.125, 0.150)	0.292	42.87 ± 0.36	54	0.68
[0.150, 0.175)	0.297	35.43 ± 0.32	43	0.83
[0.175, 0.200)	0.291	30.07 ± 0.30	36	0.96
[0.200, 0.225)	0.302	26.42 ± 0.28	32	1.13
[0.225, 0.250)	0.308	23.70 ± 0.26	30	1.28
[0.250, 0.275)	0.305	21.13 ± 0.25	32	1.42
[0.275, 0.300)	0.311	19.23 ± 0.24	29	1.59
[0.300, 0.325)	0.320	17.68 ± 0.23	35	1.78
[0.325, 0.350)	0.320	15.96 ± 0.22	23	1.97
[0.350, 0.375)	0.336	14.64 ± 0.21	19	2.24
[0.375, 0.400)	0.342	13.66 ± 0.20	18	2.44
[0.400, 0.425)	0.354	13.17 ± 0.20	25	2.62
[0.425, 0.450)	0.367	12.16 ± 0.19	16	2.93
[0.450, 0.475)	0.378	11.56 ± 0.18	20	3.16
[0.475, 0.500)	0.396	10.72 ± 0.18	15	3.57

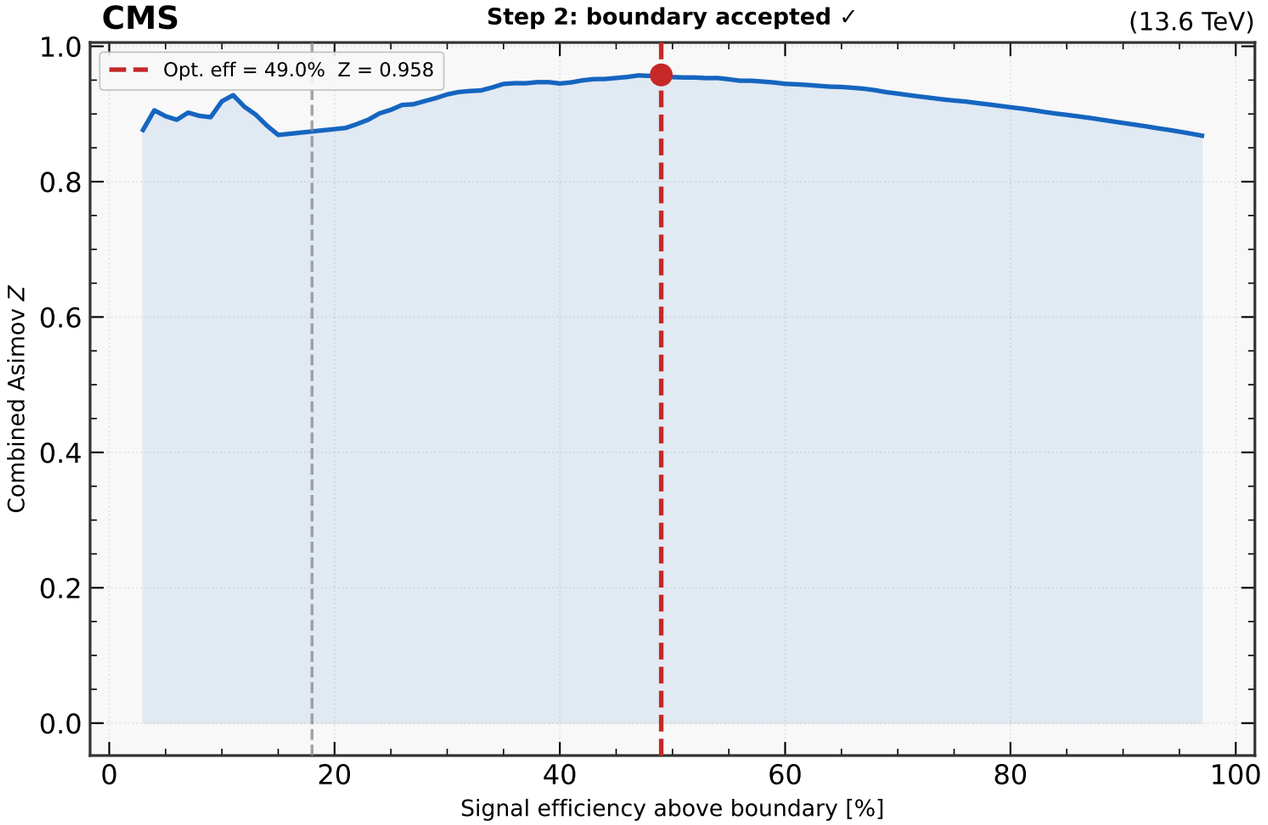
BDT bin summary: signal, background, and sensitivity (page 2 / 2)

BDT bin	Total signal	Bkg. ± ΔB	Data	S/(S+B) (%)
[0.500, 0.525)	0.406	10.22 ± 0.17	11	3.82
[0.525, 0.550)	0.436	10.01 ± 0.17	10	4.18
[0.550, 0.575)	0.455	9.56 ± 0.17	14	4.54
[0.575, 0.600)	0.491	9.26 ± 0.17	11	5.04
[0.600, 0.625)	0.524	9.04 ± 0.16	7	5.48
[0.625, 0.650)	0.564	8.59 ± 0.16	14	6.15
[0.650, 0.675)	0.602	8.41 ± 0.16	10	6.68
[0.675, 0.700)	0.651	8.01 ± 0.15	18	7.51
[0.700, 0.725)	0.717	8.10 ± 0.16	16	8.14
[0.725, 0.750)	0.797	7.50 ± 0.15	8	9.60
[0.750, 0.775)	0.886	7.53 ± 0.15	8	10.53
[0.775, 0.800)	1.001	7.15 ± 0.15	9	12.29
[0.800, 0.825)	1.146	7.27 ± 0.15	5	13.61
[0.825, 0.850)	1.326	7.02 ± 0.14	11	15.88
[0.850, 0.875)	1.552	7.02 ± 0.14	8	18.10
[0.875, 0.900)	1.880	6.60 ± 0.14	7	22.16
[0.900, 0.925)	2.374	6.10 ± 0.13	7	28.00
[0.925, 0.950)	3.203	6.05 ± 0.13	9	34.63
[0.950, 0.975)	4.907	5.16 ± 0.12	7	48.72
[0.975, 1.000)	9.744	3.00 ± 0.09	8	76.46

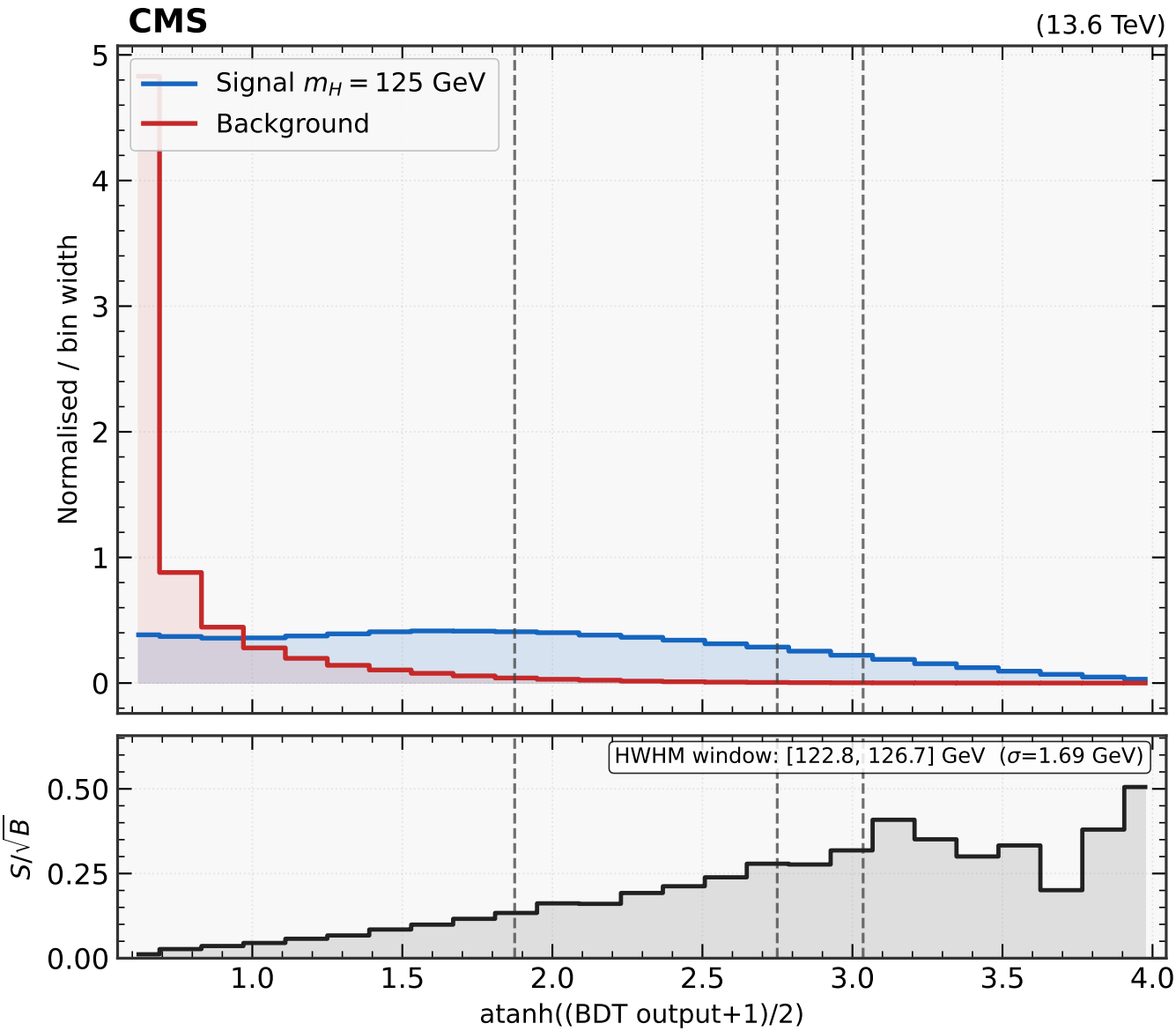

Combined significance vs. number of VBF BDT categories
CMS
(13.6 TeV)
Optimal: n=4, Z=1.0248
0.178
0.853
0.958
1.025
Combined Asimov Z [HWHM window]
Number of BDT categories
1.0
0.8
0.6
0.4
0.2
1
2
3
4

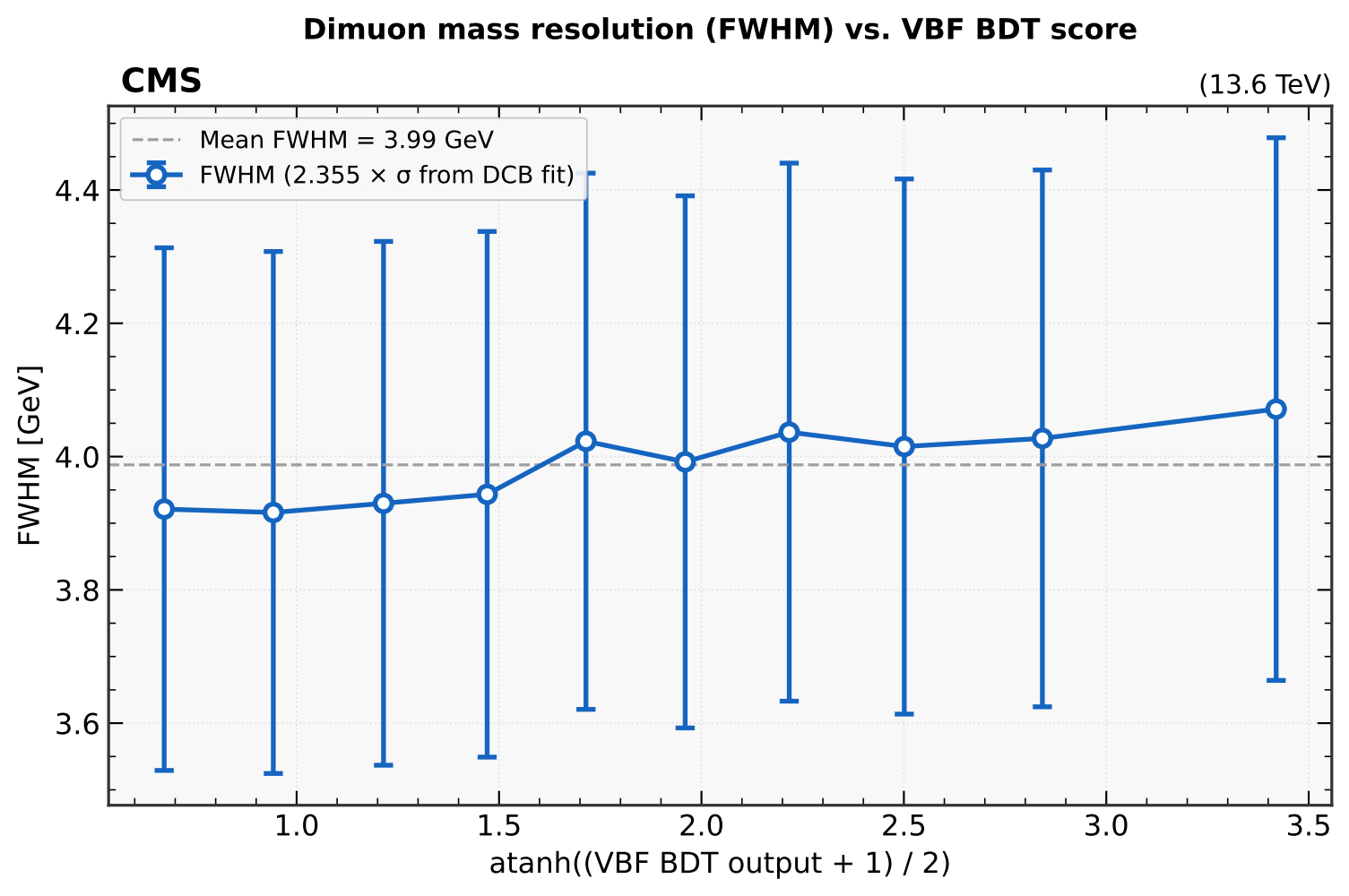
Iterative category boundary optimisation (AN Fig. 32 analog)

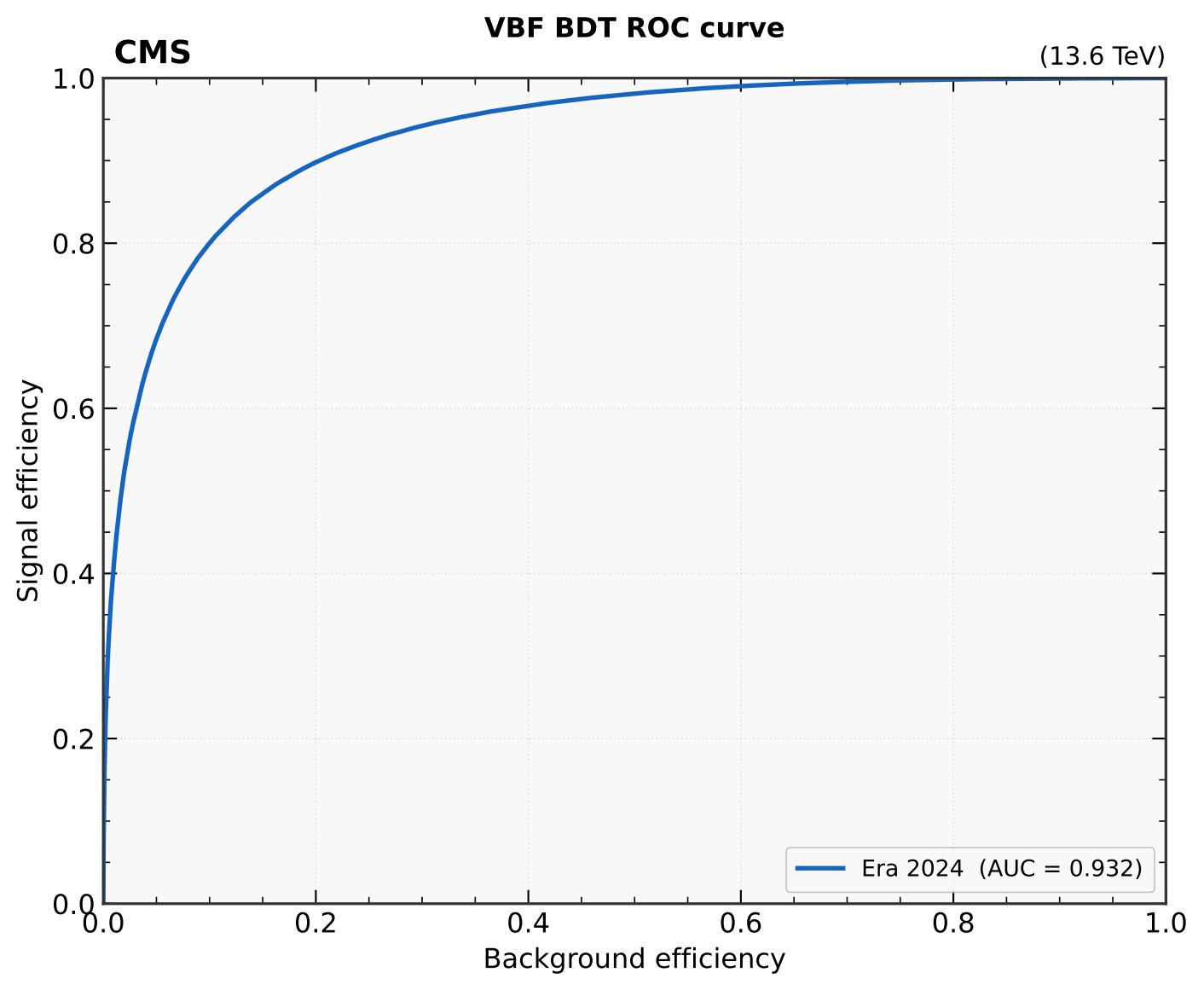


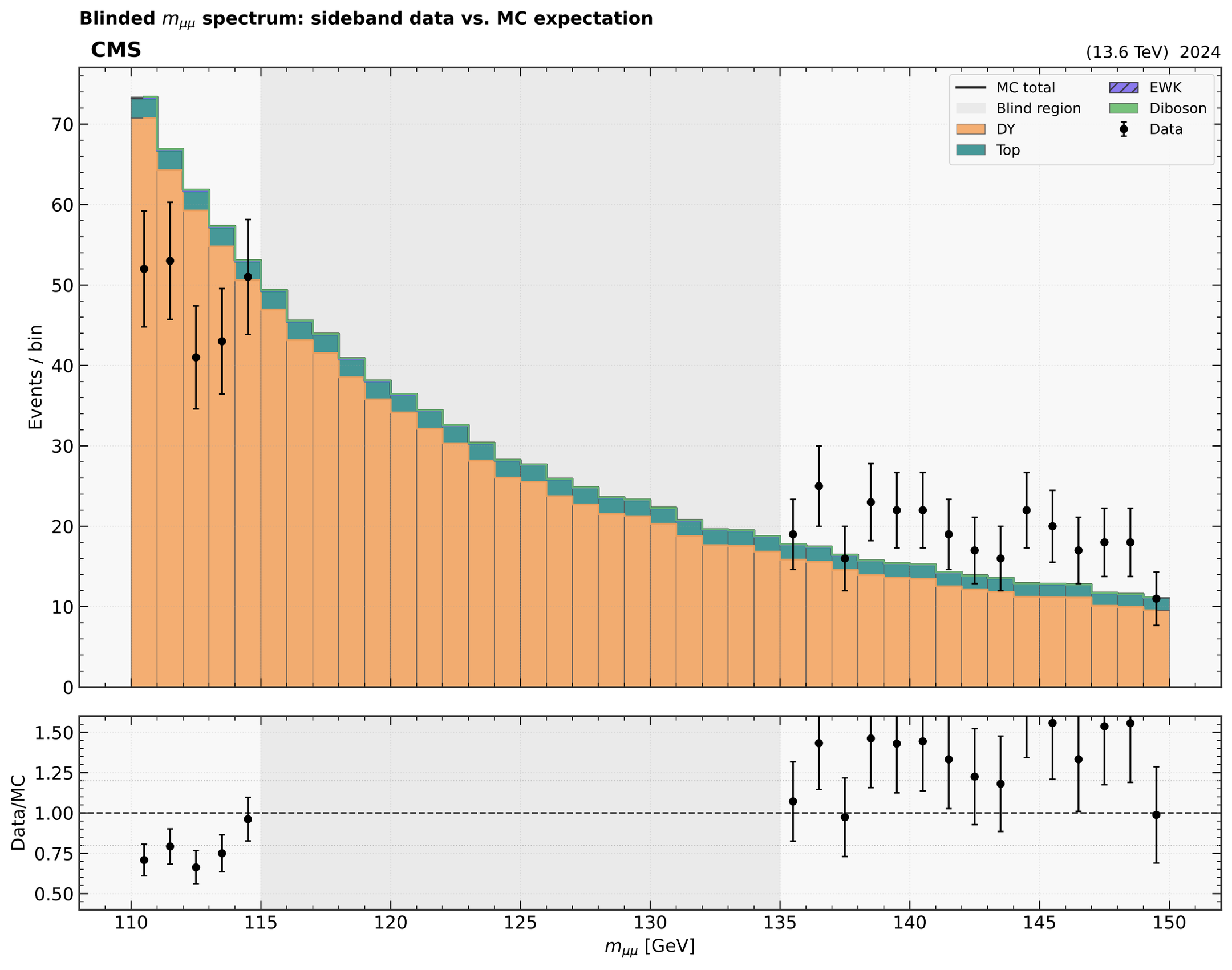


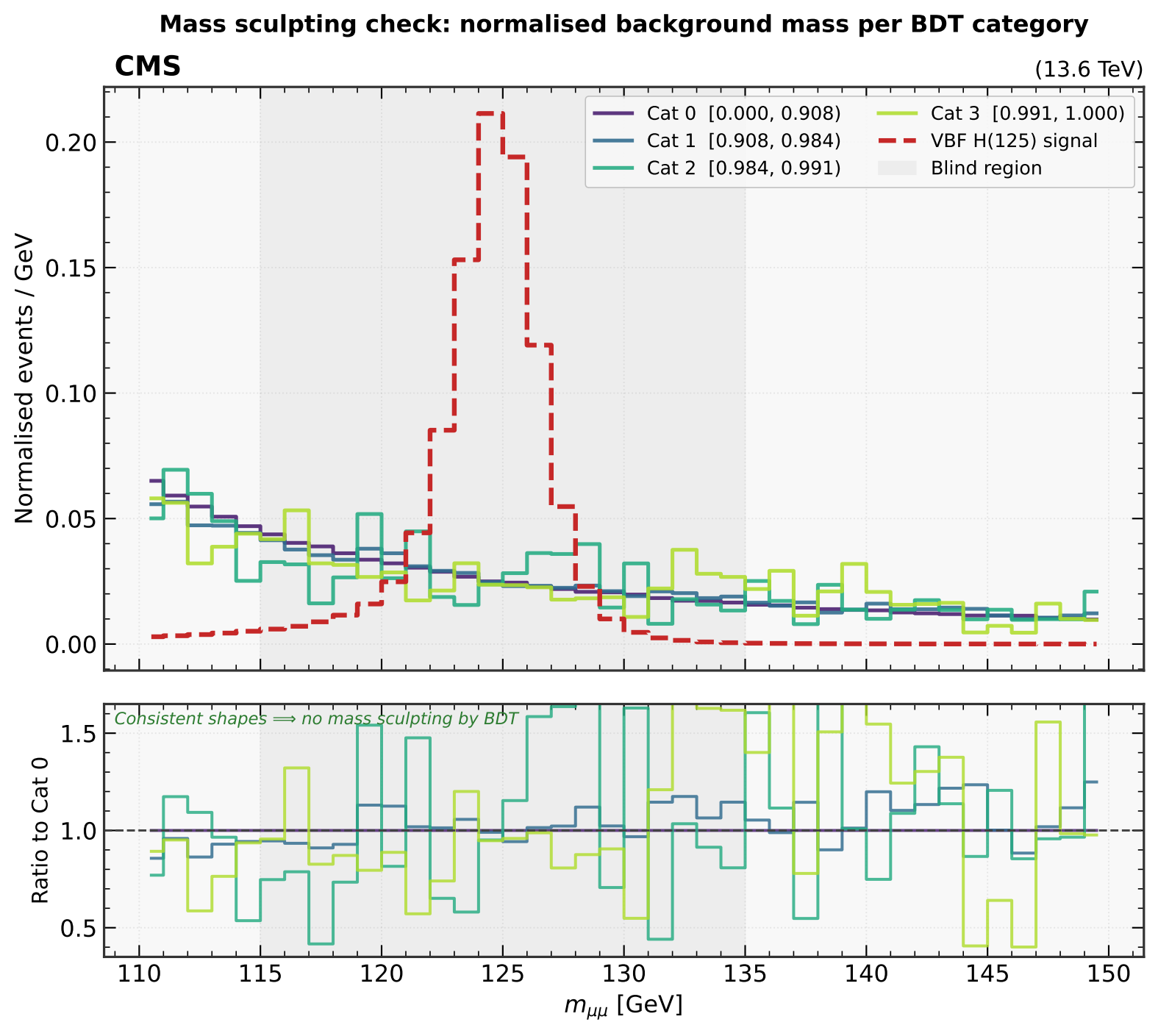
VBF BDT score and per-bin $S/\sqrt{B}$ in signal HWHM window




Dimuon mass resolution (FWHM) vs. VBF BDT score
CMS
(13.6 TeV)
Mean FWHM = 3.99 GeV
FWHM (2.355 × σ from DCB fit)
FWHM [GeV]
4.4
4.2
4.0
3.8
3.6
1.0
1.5
2.0
2.5
3.0
3.5
atanh((VBF BDT output + 1) / 2)

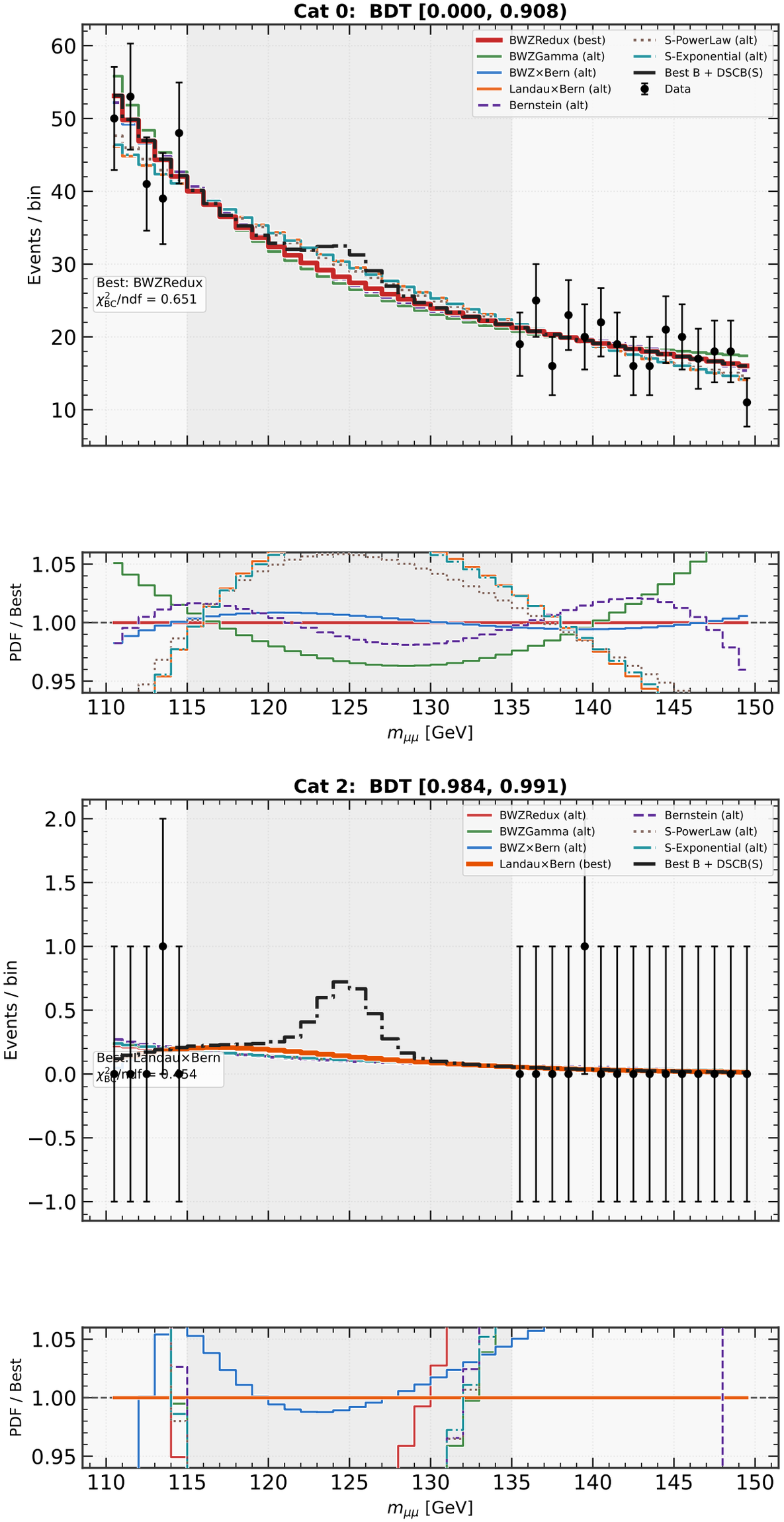

VBF BDT ROC curve
CMS
(13.6 TeV)
Signal efficiency
Background efficiency
0.0
0.2
0.4
0.6
0.8
1.0
Era 2024 (AUC = 0.932)


Blinded $m_{\mu\mu}$ spectrum: sideband data vs. MC expectation
CMS
(13.6 TeV) 2024
MC total
Blind region
DY
Top
EWK
Diboson
Data
Events / bin
Data/MC
$m_{\mu\mu}$ [GeV]

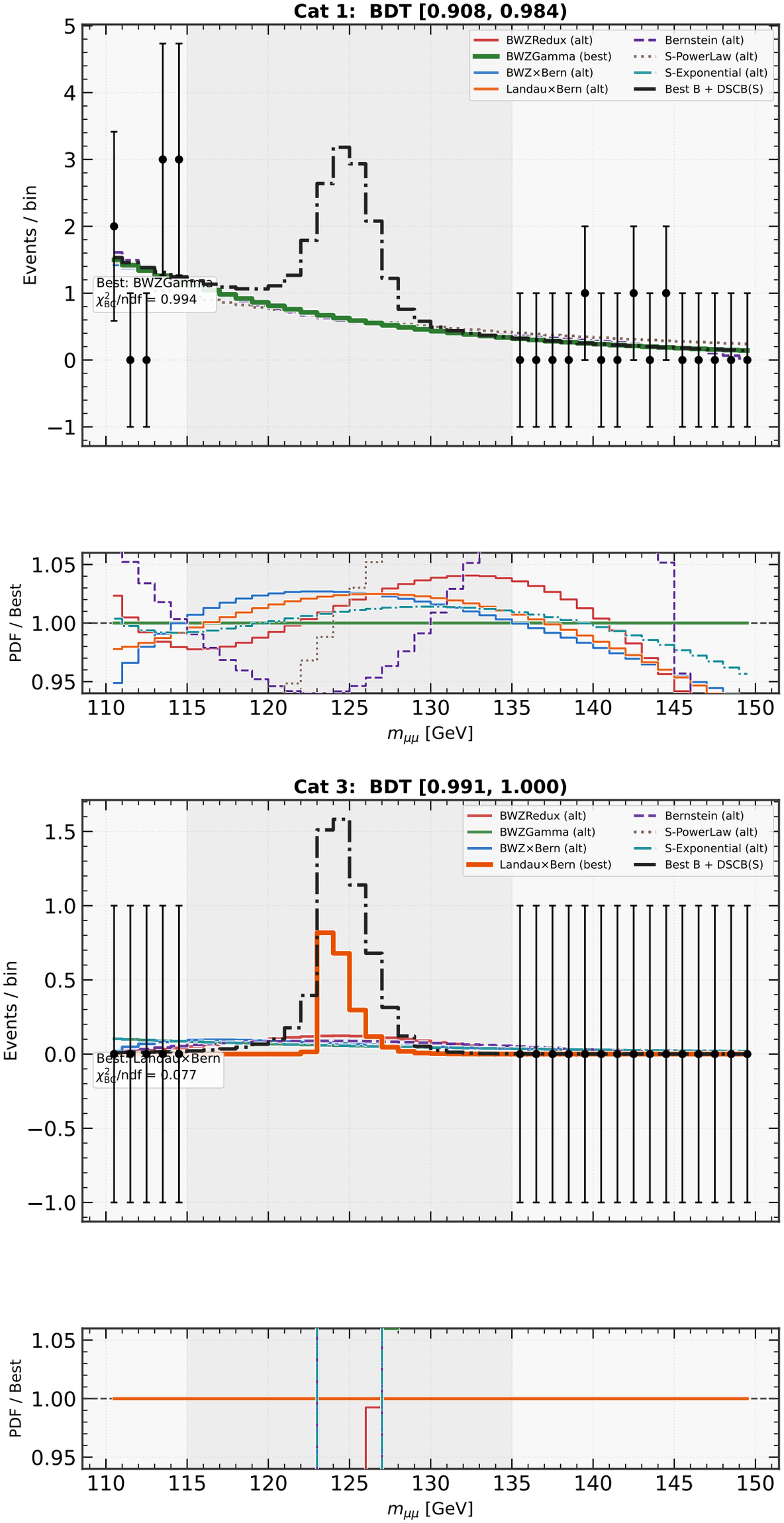

Mass sculpting check: normalised background mass per BDT category
CMS
(13.6 TeV)
Cat 0 [0.000, 0.908)
Cat 1 [0.908, 0.984)
Cat 2 [0.984, 0.991)
Cat 3 [0.991, 1.000)
VBF H(125) signal
Blind region
Normalised events / GeV
0.20
0.15
0.10
0.05
0.00
Consistent shapes ⟹ no mass sculpting by BDT
Ratio to Cat 0
1.5
1.0
0.5
110
115
120
125
130
135
140
145
150
$m_{\mu\mu}$ [GeV]

Per-category dimuon mass: background model fits

Cat 0: BDT [0.000, 0.908)



Cat 1: BDT [0.908, 0.984)



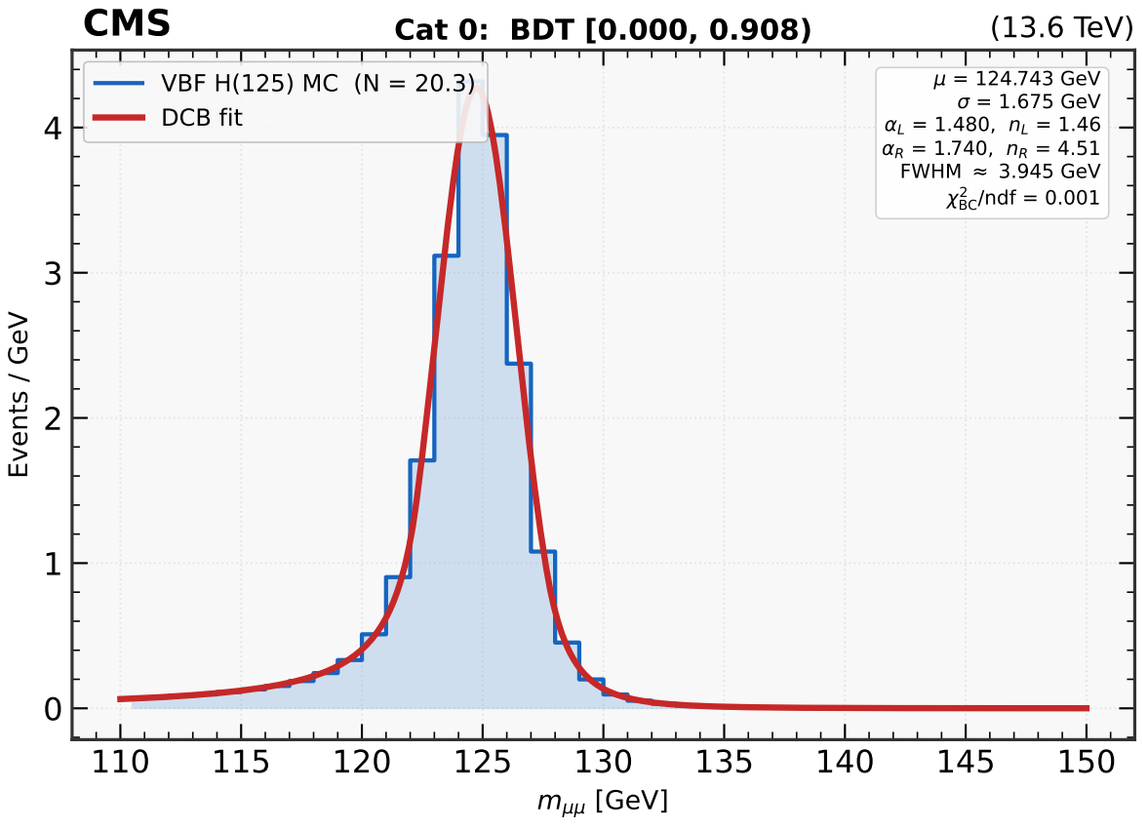
Cat 2: BDT [0.984, 0.991)

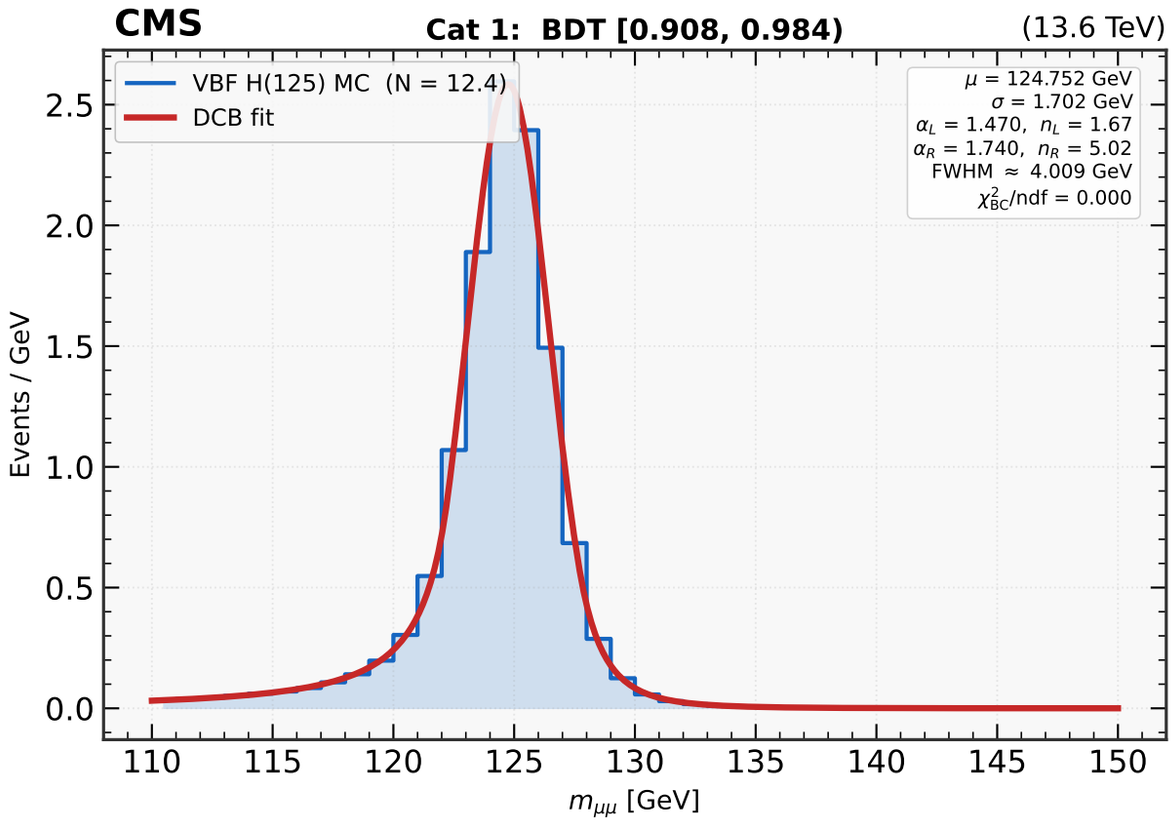
Cat 3: BDT [0.991, 1.000)

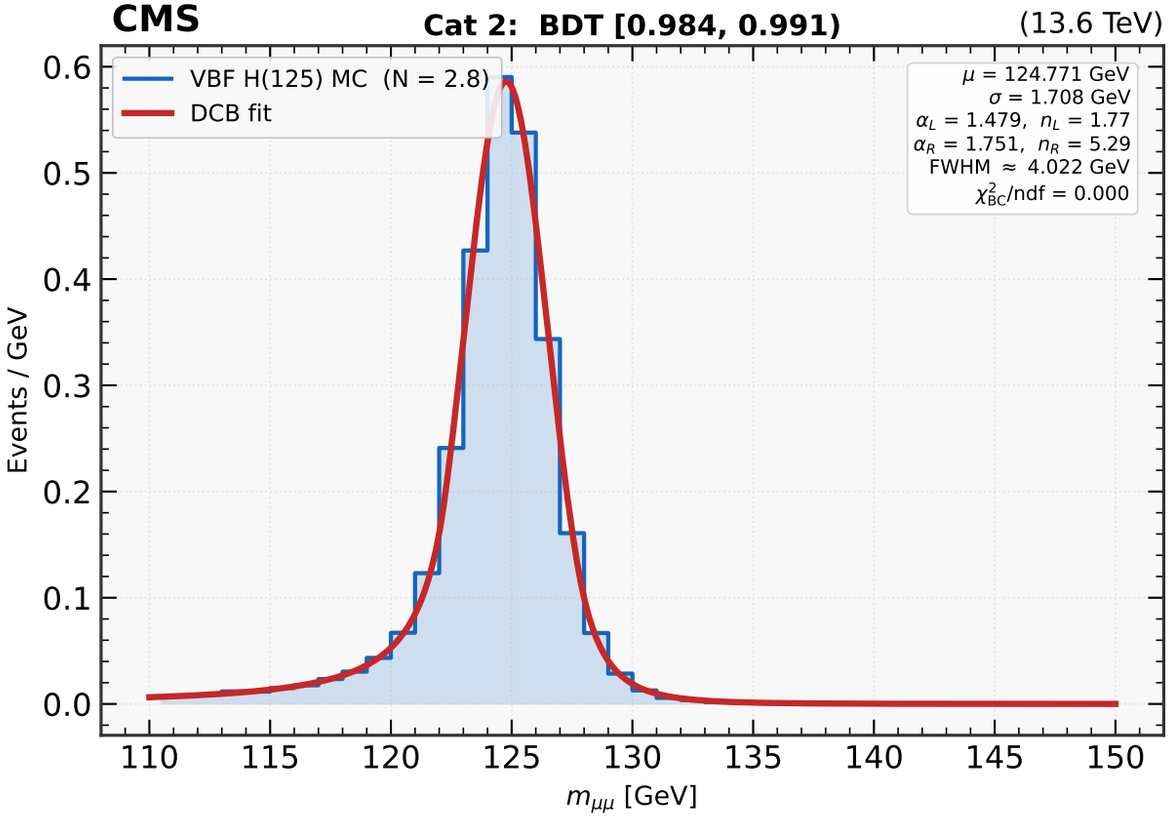
Per-category background model fit comparison (χ²_BC/ndf, ndf)

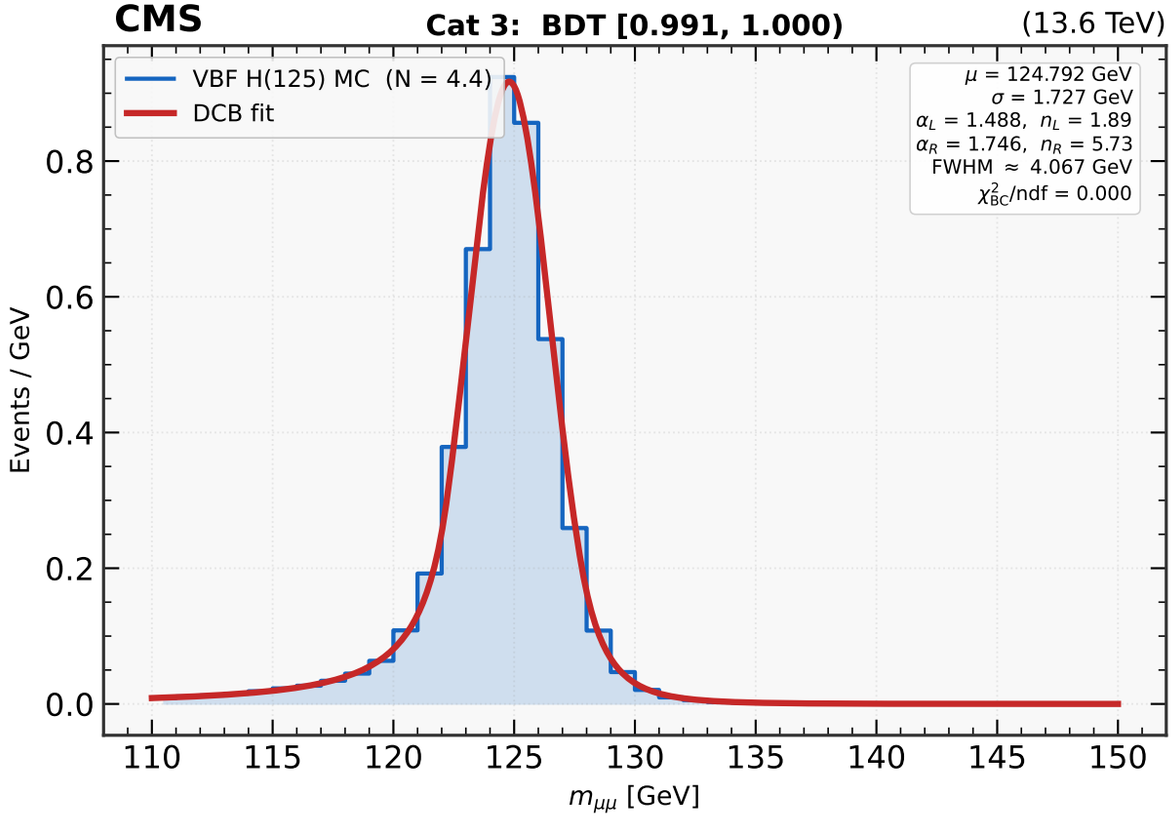
Category	BDT range	Best model	Best χ²_BC/ndf	BWZRedux	BWZGamma	BWZ×Bern	Landau×Bern	Bernstein	S-PowerLaw	S-Exponential
0	[0.000, 0.908)	BWZRedux	0.651	0.651 (37)	0.681 (37)	0.666 (36)	0.767 (37)	0.661 (35)	0.711 (37)	0.755 (37)
1	[0.908, 0.984)	BWZGamma	0.994	0.995 (37)	0.994 (37)	1.022 (36)	0.994 (37)	1.035 (35)	1.021 (37)	0.994 (37)
2	[0.984, 0.991)	Landau×Bern	0.454	0.460 (37)	0.459 (37)	0.459 (36)	0.454 (37)	0.480 (35)	0.460 (37)	0.460 (37)
3	[0.991, 1.000)	Landau×Bern	0.077	0.228 (37)	0.312 (37)	0.293 (36)	0.077 (37)	0.278 (35)	0.313 (37)	0.311 (37)

VBF H(125) signal DCB extraction per BDT category



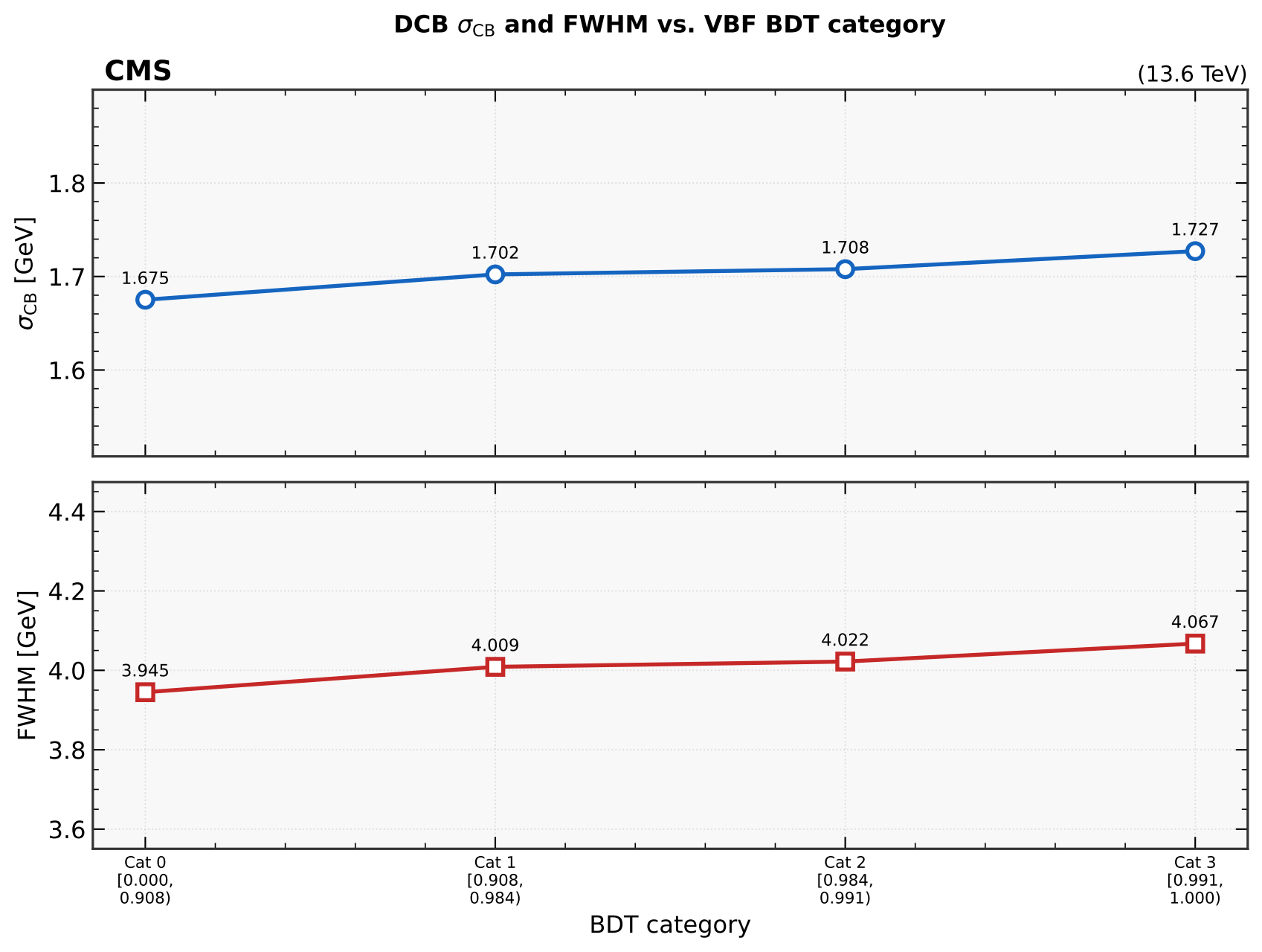


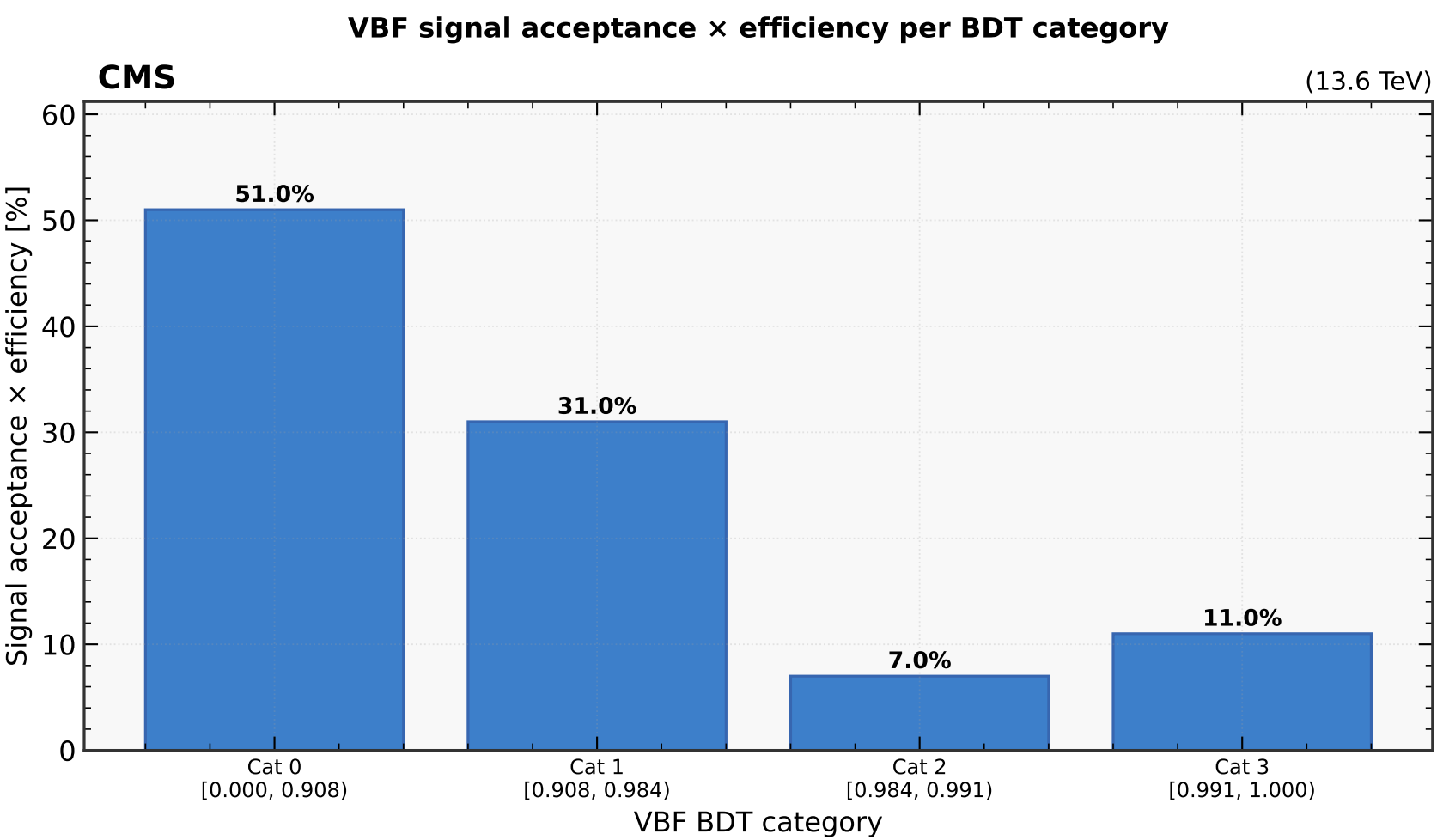




VBF signal DCB fit parameters per category

Category	BDT range	N_sig	Mean [GeV]	Sigma [GeV]	alpha_L	n_L	alpha_R	n_R	FWHM [GeV]	Asimov Z	χ^2_BC/ndf
0	[0.000, 0.908)	20.33	124.7432	1.6751	1.4801	1.461	1.7404	4.513	3.945	0.091	0.001
1	[0.908, 0.984)	12.36	124.7520	1.7022	1.4700	1.666	1.7398	5.023	4.009	0.448	0.000
2	[0.984, 0.991)	2.79	124.7715	1.7079	1.4792	1.771	1.7515	5.294	4.022	0.423	0.000
3	[0.991, 1.000)	4.39	124.7916	1.7271	1.4880	1.887	1.7463	5.734	4.067	0.821	0.000


DCB σ_{CB} and FWHM vs. VBF BDT category
CMS
(13.6 TeV)
σ_{CB} [GeV]
1.8
1.7
1.6
1.675
1.702
1.708
1.727
FWHM [GeV]
4.4
4.2
4.0
3.8
3.6
3.945
4.009
4.022
4.067
Cat 0 [0.000, 0.908)
Cat 1 [0.908, 0.984)
Cat 2 [0.984, 0.991)
Cat 3 [0.991, 1.000)
BDT category


VBF signal acceptance × efficiency per BDT category
CMS
(13.6 TeV)
Signal acceptance × efficiency [%]
0
10
20
30
40
50
60
51.0%
31.0%
7.0%
11.0%
Cat 0
[0.000, 0.908)
Cat 1
[0.908, 0.984)
Cat 2
[0.984, 0.991)
Cat 3
[0.991, 1.000)
VBF BDT category

Per-category yield and sensitivity summary

Category	BDT range	S_full	B_full	S_HWHM	B_HWHM	HWHM window [GeV]	S/√B	Asimov Z
0	[0.000, 0.908)	20.33	227027.79	13.740	22608.766	[122.8, 126.7]	0.091	0.091
1	[0.908, 0.984)	12.36	3420.72	8.430	350.900	[122.7, 126.8]	0.450	0.448
2	[0.984, 0.991)	2.79	190.83	1.915	19.847	[122.8, 126.8]	0.430	0.423
3	[0.991, 1.000)	4.39	122.71	3.036	12.697	[122.8, 126.8]	0.852	0.821

Category summary: HWHM-windowed signal and background yields

Category	BDT range	Data sidebands	MC sidebands ± stat	Data/MC (sidebands)	MC full (110-150) ± stat	S/√B [HWHM]	Asimov Z (cat) [HWHM]	B_HWHM
0	[0.000, 0.908)	512	105420.56 ± 255.39	0.005	227027.79 ± 375.00	0.0914	0.0914	22608.766
1	[0.908, 0.984)	11	1552.40 ± 30.40	0.007	3420.72 ± 45.21	0.4500	0.4483	350.900
2	[0.984, 0.991)	2	89.66 ± 7.00	0.022	190.83 ± 10.17	0.4299	0.4233	19.847
3	[0.991, 1.000)	0	57.12 ± 5.18	0.000	122.71 ± 7.62	0.8519	0.8210	12.697
Combined	all						1.0307	

Category composition by process (full mass window)

Category	BDT range	DY	EWK	Top	Diboson	total_err2	total
0	[0.000, 0.908)	210689.80	237.65	15708.00	392.33	140623.13	227027.79
1	[0.908, 0.984)	3060.37	122.35	226.98	11.02	2044.06	3420.72
2	[0.984, 0.991)	154.45	25.06	10.76	0.56	103.37	190.83
3	[0.991, 1.000)	86.55	30.71	5.23	0.22	58.11	122.71